

# DASE7506 Project 1 Language Model BPB Technical Report

*Frozen v3: driving the complete test BPB from 2.101265 to 1.571349*

## 1. Background and Mission Objectives

A language model estimates the probability of the next token from the preceding context. The higher the probability it assigns to the correct token, the fewer bits are needed to encode the same text. This project therefore cares not only about whether the model fits the training text, but also about whether it produces accurate and reproducible probability distributions on text it has not been trained on.

The task is to train a decoder-only Transformer from random initialisation on the supplied WikiText-2 text and to minimise bits per byte (BPB) on the complete test split under a fixed BPE-2048 tokenizer, a 256-token context and the supplied CPU FP32 scorer. Only the training split updates parameters; architecture, hyperparameters, seeds and checkpoint selection are decided on validation alone; the test split is scored once, after the method is frozen.

This round builds on the frozen v2 predictor (test BPB 1.611178). The frozen v3 reaches 1.571349, 25.22% below the course baseline and 0.039830 below v2.

**Table 1.** Fixed conditions and rigid limits

| Item           | Fixed setting                 | Role or limit                                        |
|----------------|-------------------------------|------------------------------------------------------|
| Data           | Supplied WikiText-2           | Train, validation and test splits strictly separated |
| Tokenizer      | BPE-2048                      | Vocabulary and segmentation are not modifiable       |
| Context        | 256 tokens                    | Training and evaluation share the maximum context    |
| Windows        | Independent, strictly causal  | No future tokens and no cross-window state           |
| Model choice   | Validation only               | Test never selects structure or hyperparameters      |
| Final scoring  | CPU FP32, complete test split | The reported score reproduces on a CPU               |
| CPU time limit | At most 5x the baseline       | Hard limit; a lower BPB cannot offset it             |
| Memory limit   | Peak RSS at most 4 GiB        | Hard limit                                           |
| Asset limit    | At most 64 MiB uncompressed   | Hard limit                                           |

## 2 Evaluation Indicators and Selection Agreement

The headline metric is bits per byte. With  $B$  the raw UTF-8 byte count of the split, the total negative log-likelihood is converted to bits through  $\ln(2)$  and divided by  $B$ . Lower is better. A complete split must accumulate all target NLL first and divide once, never averaging the BPB of individual batches or windows.

**Table 2.** Reported metrics and assessment

| Metric           | Definition or measurement                 | Role                        | Direction or limit |
|------------------|-------------------------------------------|-----------------------------|--------------------|
| BPB              | total NLL / $\ln(2)$ / UTF-8 bytes        | headline quality            | lower better       |
| Token PPL        | $\exp(\text{total NLL} / \text{targets})$ | token-level difficulty      | lower better       |
| CPU time ratio   | candidate time / same-split baseline time | inference speed cost        | at most 5x         |
| Peak RSS         | maximum resident memory of the scorer     | evaluation memory           | at most 4 GiB      |
| Inference assets | uncompressed assets needed at inference   | checkpoint and dependencies | at most 64 MiB     |

BPB is the ranking metric; the three resource limits are hard constraints that stand on their own and cannot be offset by a lower BPB. Quality and resources are therefore reported separately, and resources are re-measured with a median of five repetitions before freezing.

## 2.1 Data usage and freezing

Train updates parameters, validation selects structure, hyperparameters, seeds and checkpoints, and test is scored once after the freeze manifest has been written. The manifest records the SHA-256 of the checkpoint, the implementation files, the evaluator and the tokenizer, and it leaves the test-result field empty to show the freeze preceded any test number.

Comparisons must also hold the scorer, the context length and the number of processed targets fixed. Where training scale or duration changes, quality and cost are reported separately, so that "trained more" or "model larger" is never written up as a single mechanism.

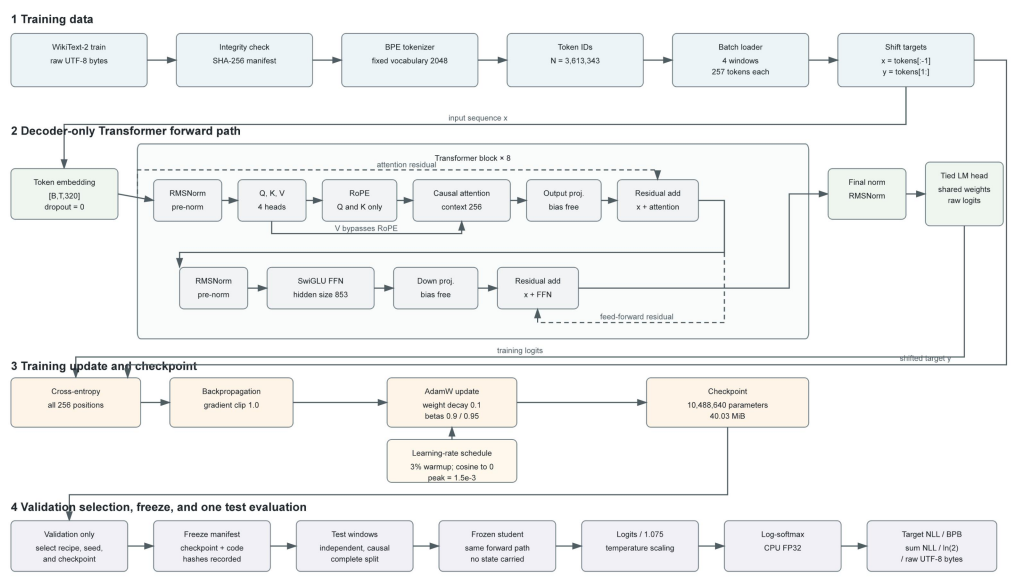
**Table 3.** Scale and training budget of this round

| Item                | Value                            | Note                                       |
|---------------------|----------------------------------|--------------------------------------------|
| Training corpus     | 3,613,343 BPE tokens             | about 3.6M; the dominant data limit here   |
| Screening budget    | 9.83M to 28.9M processed targets | about 2.7 to 8.0 epochs                    |
| Final recipe budget | 28,909,568 targets               | batch 4, 28,232 optimisation steps         |
| Controlled runs     | 32 runs this round               | plus 33 runs during v1 and v2              |
| Search cost         | 773,704,448 targets              | training size is unlimited, only disclosed |
| Final parameters    | 10,488,640                       | identical to v2                            |

## 3 Models and Training Methods

The model is a pre-norm decoder-only Transformer: width 320, depth 8, four attention heads (head dimension 80), RMSNorm, a SwiGLU feed-forward with hidden size 853 (FLOP-matched to a 4x GELU MLP), RoPE positional encoding, and tied input and output weights, for a total of 10,488,640 parameters.

Training uses AdamW (weight decay 0.1, betas 0.9/0.95), gradient clipping at 1.0, a three percent warmup followed by cosine decay, and without-replacement window sampling over every legal window start. Relative to v2, v3 changes the seed, batch size, peak learning rate, decay floor and checkpoint location together, then divides the logits by a validation-fitted temperature of 1.075; only the before-and-after temperature comparison is assigned a separate causal interpretation, while the other changes are reported as a joint recipe difference.



**Fig. 1.** Model structure, training data flow and evaluation path of frozen v3. The annotated batch size, learning rate and decay floor are the v3 values.

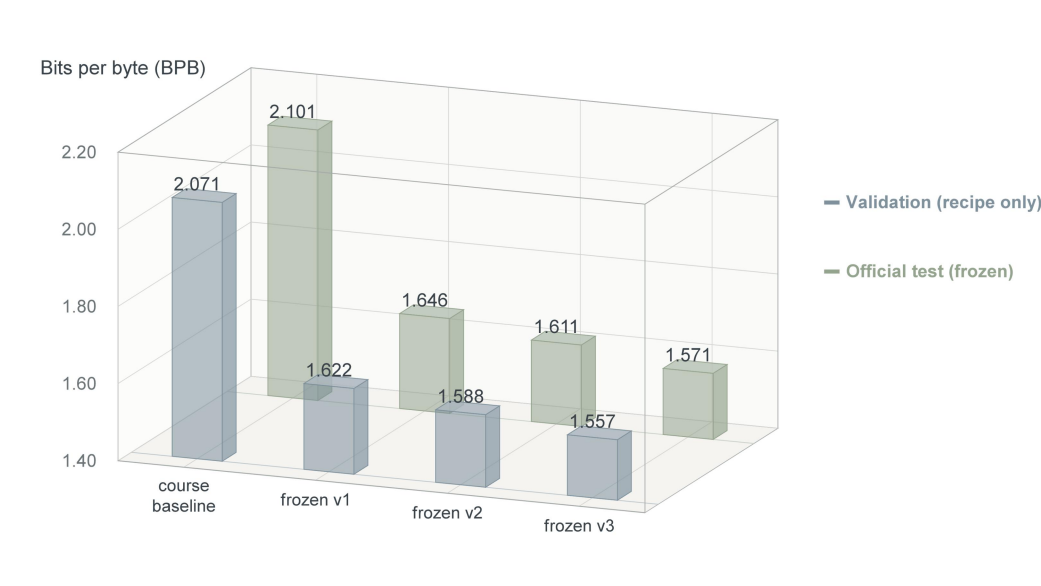
**Table 4.** Model and training recipe: v2 against v3

| Setting                 | v2                      | v3                      | Effect on inference             |
|-------------------------|-------------------------|-------------------------|---------------------------------|
| Width / depth / heads   | 320 / 8 / 4             | 320 / 8 / 4             | none                            |
| Norm / FFN / positional | RMSNorm / SwiGLU / RoPE | RMSNorm / SwiGLU / RoPE | none                            |
| Weight tying            | tied                    | tied                    | none                            |
| Parameters              | 10,488,640              | 10,488,640              | none                            |
| Batch (windows)         | 32                      | 4                       | none, training only             |
| Peak learning rate      | 1e-3                    | 1.5e-3                  | none, training only             |
| Decay floor             | 0.1 x peak              | 0                       | none, training only             |
| Calibration temperature | 1.00                    | 1.075                   | one elementwise scalar division |
| Inference assets        | 40.033 MiB              | 40.034 MiB              | +1 KB                           |

Batch size, learning rate and the decay floor affect training only, while temperature scaling adds one elementwise division to the inference graph without adding trainable parameters; the asset increase is about 1 KB and the operation is negligible beside the Transformer forward pass.

## 4 BPB Optimization Process and Judgment Methods

v1 used learned positional embeddings and froze RMSNorm, SwiGLU, tied weights and the trainer changes together; v2 introduced RoPE and selected among predeclared validation checkpoints during longer training; v3 keeps a 28.9M-target budget, changes the optimisation and checkpoint recipe jointly, and calibrates the output confidence on validation.



**Fig. 2.** Validation and official test BPB of the four versions. Points come from independent runs; the v3 validation bar is the uncalibrated recipe score, and the frozen configuration additionally applies  $T = 1.075$ .

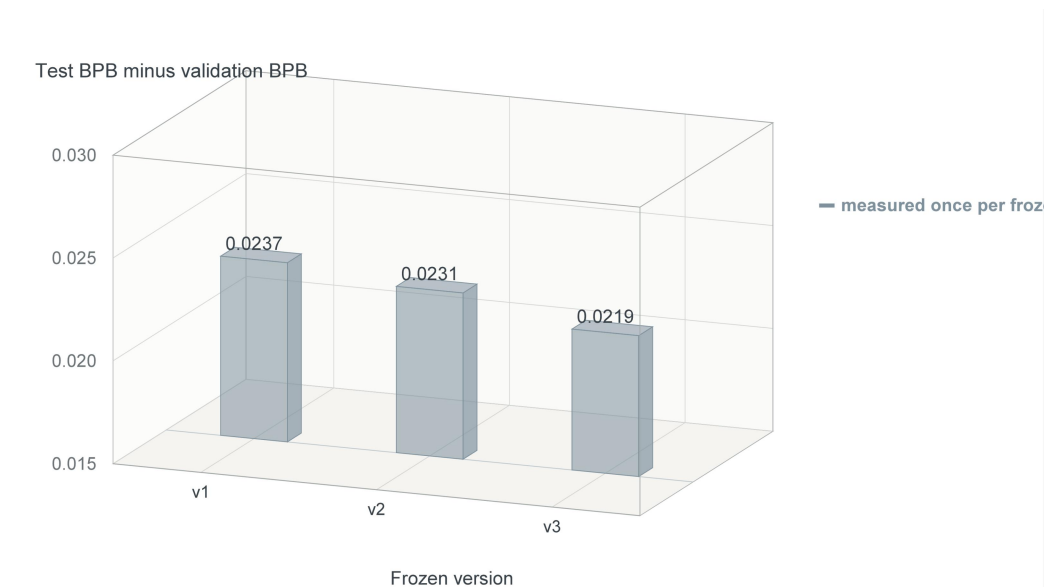
### 4.1 Numerical decomposition of the optimisation path

v3 lowers test BPB by 0.039830 relative to v2. On validation, the frozen v2 setting and the uncalibrated v3 setting differ by 0.031205, but that contrast combines the seed, batch, learning rate, decay floor and checkpoint location and is therefore a joint recipe gap rather than a causal contribution; the paired temperature comparison on the same v3 checkpoint gives a separately attributable calibration gain of 0.007392. These validation differences cannot be added and equated with the test delta. Earlier, v2 improved on v1 by 0.034974 and v1 on the baseline by 0.455113.

**Table 5.** Stage-to-stage deltas and their interpretation limits

| Stage                         | Validation BPB | Test BPB | Delta    | Interpretation limit                                 |
|-------------------------------|----------------|----------|----------|------------------------------------------------------|
| Course baseline               | 2.071087       | 2.101265 | —        | reference implementation                             |
| v1 structure + trainer        | 1.622441       | 1.646152 | 0.455113 | structure, trainer and budget all change             |
| v2 long run + selection       | 1.588095       | 1.611178 | 0.034974 | duration and checkpoint rule both change             |
| v3 joint recipe + calibration | 1.556890       | 1.571349 | 0.039830 | seed, optimisation, selection and calibration change |

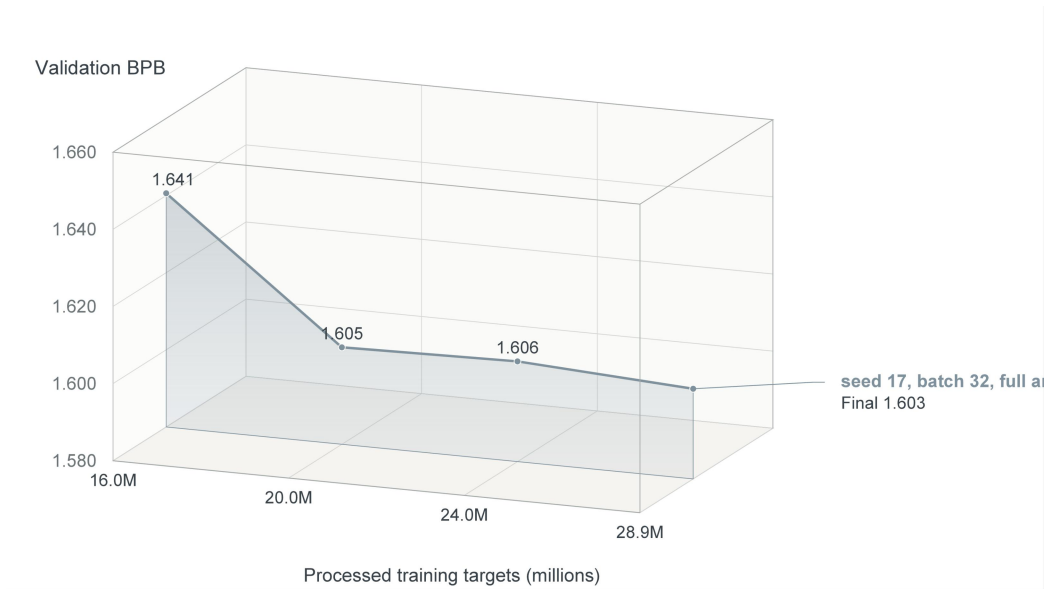
The path to v2 narrowed the search space in recorded steps. A 9.83M-target 2 by 2 factorial gave a SwiGLU main effect of -0.055444 and an RMSNorm main effect of -0.005725; width 320, depth 8 and a 1e-3 peak rate then became the capacity reference, while dropout 0.05 and weight decay 0.2 produced no reproducible gain. After the v1 long run with learned positions, short RoPE controls improved both tested seeds, whereas untying, larger capacity, EMA and multi-token prediction were rejected by resource or noise evidence. The first long RoPE run regressed after 19.66M targets, so three seed runs declared checkpoints at 9.83M, 19.66M and 28.91M in advance and selected the lowest complete-validation score; seed 137 at 19.66M was frozen and produced the one-shot v2 test BPB of 1.611178.



**Fig. 3.** The "test BPB minus validation BPB" offset for the three frozen versions. Three independent measurements give +0.023711, +0.023083 and +0.021850, so validation predicts test reliably and the test split never has to be scored repeatedly.

## 5 Basis for Decisions on Hyperparameters and Training Length

This section answers three questions: where the training budget should stop, what batch size and learning rate should be used, and how large the noise is before any of these choices can be judged. All comparisons use seed 17 and one scorer.



**Fig. 4.** Sweeping the training budget near 20M. The points at 20.0M, 24.0M and 28.9M are indistinguishable within noise, so the horizon saturates around 20M, while 16M is clearly short.

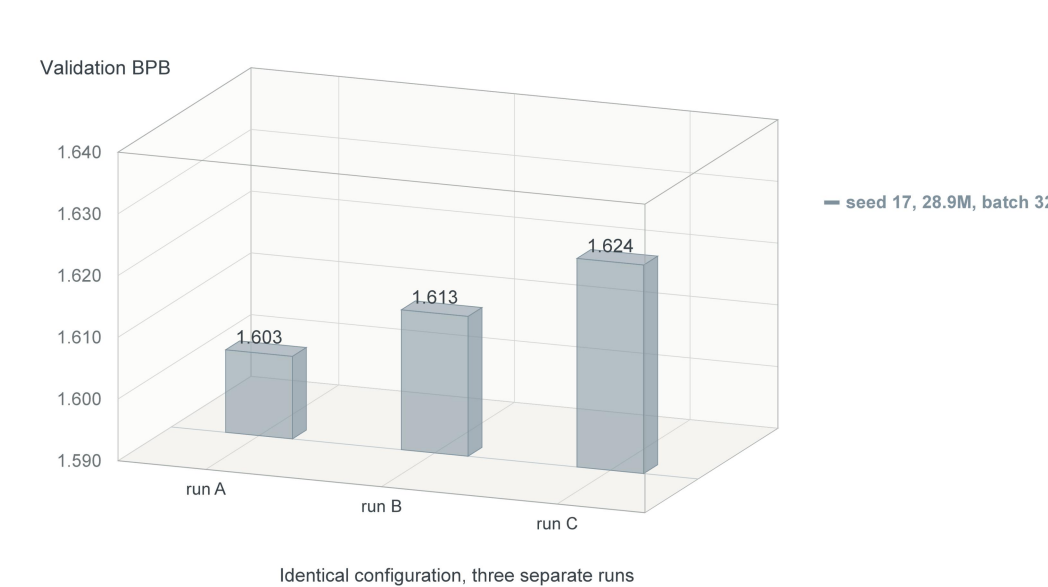
**Table 6.** Training-budget sweep (seed 17, batch 32, full anneal)

| Processed targets | Validation BPB | Relative to 20M | Verdict      |
|-------------------|----------------|-----------------|--------------|
| 16.0M             | 1.640587       | +0.035453       | insufficient |
| 20.0M             | 1.605134       | +0.000000       | saturated    |
| 24.0M             | 1.605983       | +0.000848       | saturated    |
| 28.9M             | 1.603377       | -0.001757       | saturated    |

This closes the ordinary route of training longer. Under a full anneal, pushing the budget from 20M to 28.9M buys only a noise-level difference, so the remaining improvement had to come from spending the same token budget more effectively.

### 5.1 Why training loss and validation must be read together

Training turned out not to be reproducible. The MPS bf16 kernels used for multiplication and reduction are non-deterministic and two runs of an identical configuration diverge in training loss from step 500 onwards. We therefore replicated one configuration three times to obtain a usable noise floor, and graded every later comparison against it.



**Fig. 5.** Validation BPB of three independent runs of one configuration (seed 17, 28.9M targets, batch 32, full anneal). Mean 1.613278, standard deviation 0.010223, range 0.020418. The lowest run happens to be the one previously used as the baseline.

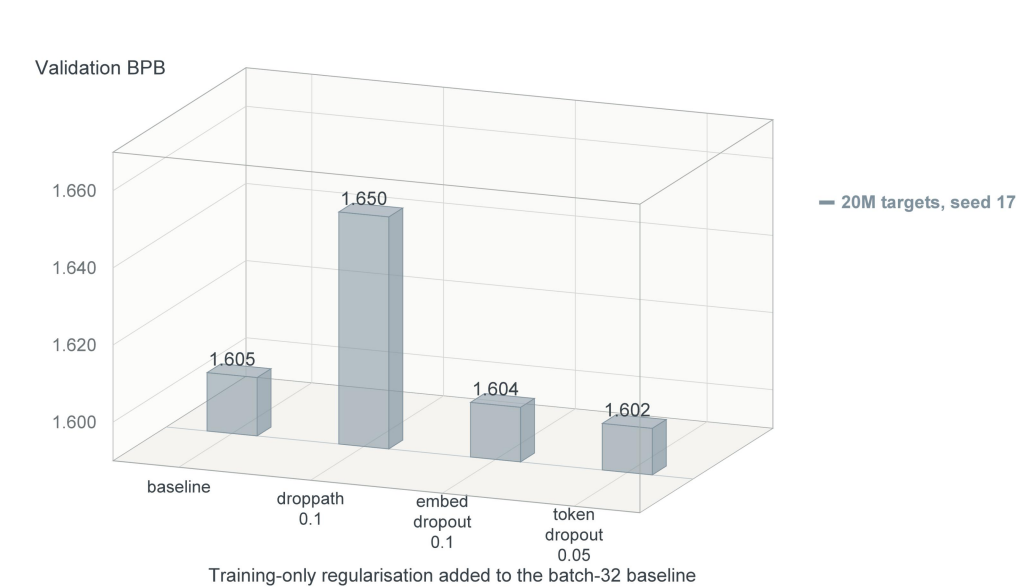
**Table 7.** Noise floor and what it implies

| Quantity                       | Value                               | Consequence                                   |
|--------------------------------|-------------------------------------|-----------------------------------------------|
| Mean of three runs             | 1.613278                            | the honest baseline for batch 32              |
| Between-run standard deviation | 0.010223                            | a single comparison must exceed about 2 sigma |
| Range of three runs            | 0.020418                            | about 2 sigma, as expected                    |
| Adoption criterion             | monotone trend + paired replication | no conclusions from single points             |

This floor changed two earlier conclusions. The claimed overfitting turning point at 19.66M rests on a +0.015 rebound that sits inside the noise; the correct statement is a plateau around 20M. The claim that "1e-3 is the best learning rate" holds only at batch 32, as section 5.3 shows. Every mechanism adopted below is therefore supported by a paired comparison or a monotone trend.

## 5.2 Scope of application for regularisation results

To confirm that the gain comes from the recipe rather than from regularisation strength, three training-time regularisers were tested against the fixed batch-32 baseline of 1.605134 at 20M targets. All were negative or unresolved. Droppath 0.1 degrades validation BPB by 0.045020, far beyond noise; whole-token dropout and embedding dropout gain only 0.003067 and 0.000981, less than a third of the noise floor.



**Fig. 6.** Validation BPB after adding training-time regularisation, against the batch-32 baseline of 1.605134 at 20M targets. Droppath 0.1 is clearly harmful; the other two sit inside the noise.

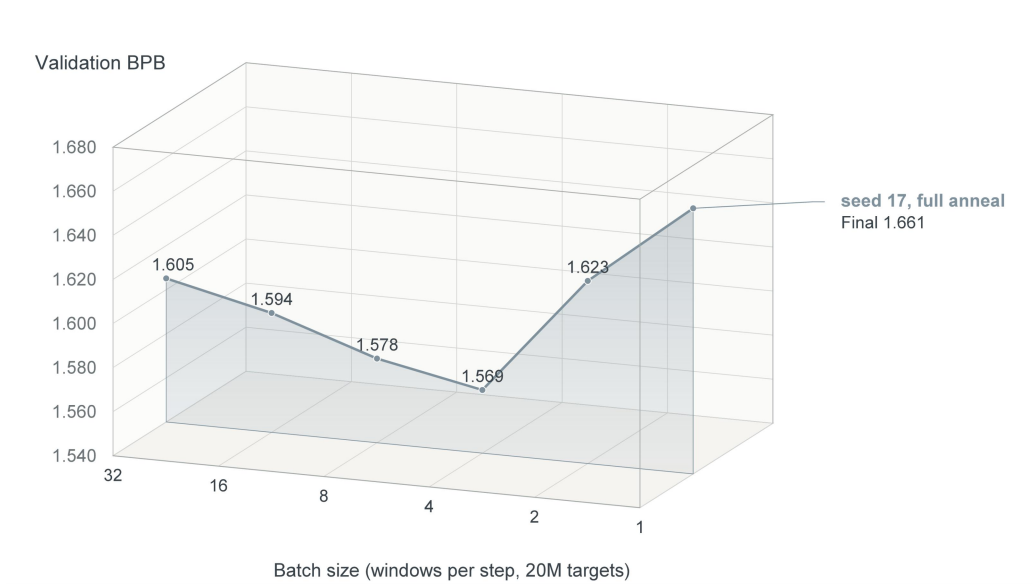
**Table 8.** Regularisation ablation (seed 17, 20M targets, batch 32)

| Direction                  | Validation BPB | Relative  | Verdict    |
|----------------------------|----------------|-----------|------------|
| baseline, no regulariser   | 1.605134       | —         | reference  |
| droppath 0.1               | 1.650154       | +0.045020 | harmful    |
| token dropout 0.05         | 1.602067       | -0.003067 | unresolved |
| embedding dropout 0.1      | 1.604153       | -0.000981 | unresolved |
| EMA 0.999 weight averaging | 1.610504       | +0.005370 | harmful    |

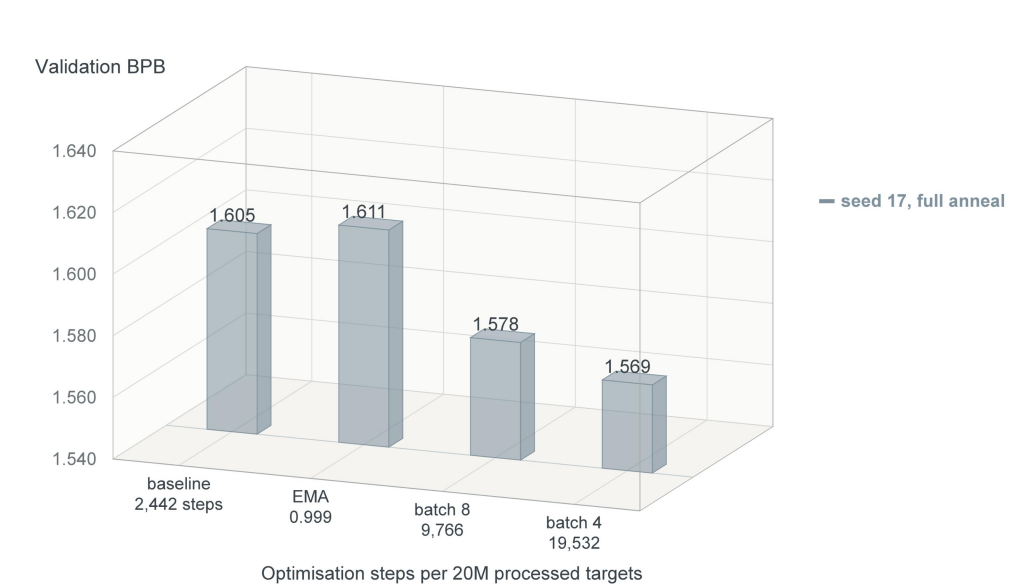
Read together with the capacity results in section 5.4, this gives the central judgement of the project: the model is neither under-regularised nor under-sized, but limited by the signal available in 3.6M training tokens. That is why adding regularisation and adding width both fail.

### 5.3 Joint choice of learning rate and batch size

This is the largest and most easily missed result of the round. At a fixed 20M targets, dropping the batch from 32 windows to 4 keeps the token budget and raises the number of optimisation steps from 2,442 to 19,532, moving validation BPB from 1.605134 to 1.568646. Continuing to 2 and 1 rebounds to 1.622860 and 1.660514, so the optimum is 4 rather than as small as possible.



**Fig. 7.** Validation BPB against batch size. The curve bottoms out at four windows and degrades on both sides: smaller batches carry noisier gradients and going too far hurts the final quality.



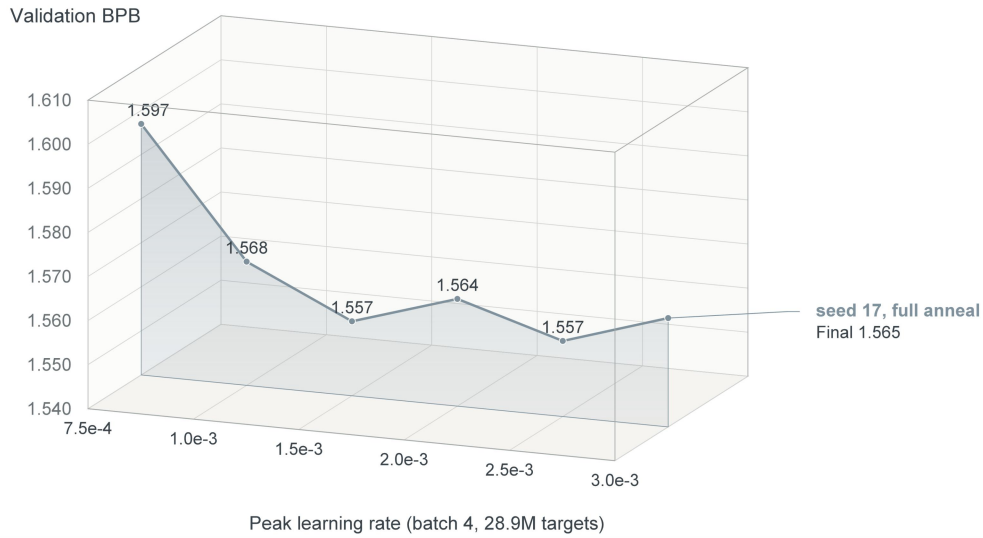
**Fig. 8.** The same results re-ordered by optimisation step count. More steps means lower final validation BPB up to 19,532 steps.

**Table 9.** Batch size and optimisation steps (seed 17, 20M targets, full anneal)

| Batch | Steps  | Validation BPB | Relative to batch 32 |
|-------|--------|----------------|----------------------|
| 32    | 2,442  | 1.605134       | +0.000000            |
| 16    | 4,883  | 1.594155       | -0.010979            |
| 8     | 9,766  | 1.578192       | -0.026942            |
| 4     | 19,532 | 1.568646       | -0.036488            |
| 2     | 39,063 | 1.622860       | +0.017726            |
| 1     | 78,125 | 1.660514       | +0.055380            |

At the common 20M-target budget, seed 17 and full anneal, validation BPB moves monotonically from 1.605134 to 1.582448 and 1.568646 as batch size falls from 32 to 8 and 4. The three batch-32 replications were run at 28.91M targets and can estimate run-to-run variability, but they cannot form a valid significance test against the single 20M-target batch-4 result; the earlier p-value is therefore removed, and the batch result is reported as a trend that still requires matched-budget replication.

Changing the batch invalidates the learning-rate sweep, and that is the most transferable lesson of the round. The rate of 1e-3 had been tuned at batch 32, where a step processes 8,192 tokens. At batch 4 a step processes 1,024 tokens, the gradient noise per step is far larger, and the optimum shifts upward.



**Fig. 9.** Re-sweeping the peak learning rate after the batch was changed to 4; six points span 7.5e-4 to 3e-3. The single runs at 1.5e-3 and 2.5e-3 differ by 0.000259, while the intervening 2e-3 point and the 3e-3 point are about 0.0075 worse and lowering the rate is clearly harmful.

**Table 10.** Learning-rate re-sweep (seed 17, batch 4, 28.9M targets)

| Peak learning rate | Validation BPB | Relative to 1e-3 | Verdict           |
|--------------------|----------------|------------------|-------------------|
| 7.5e-4             | 1.596950       | +0.028882        | too small         |
| 1.0e-3             | 1.568068       | +0.000000        | reference         |
| 1.5e-3             | 1.556890       | -0.011178        | strong candidate  |
| 2.0e-3             | 1.564327       | -0.003742        | weaker single run |
| 2.5e-3             | 1.557149       | -0.010919        | strong candidate  |
| 3.0e-3             | 1.564697       | -0.003371        | weaker single run |

Three conclusions follow. The earlier sweep stating that 1e-3 is best and 2e-3 clearly worse holds only at batch 32 and does not extrapolate; changing the batch invalidates the sweep, which is the most transferable lesson of the round. Second, extending the sweep to 3e-3 locates the two separated candidates at 1.5e-3 and 2.5e-3 differ by only 0.000259, but the worse 2e-3 result between them rules out describing the interval as a continuous plateau. With a run-to-run standard deviation near 0.010, the evidence supports 1.5e-3 as a sound frozen choice while leaving the precise learning-rate optimum unresolved pending paired replications.

A further cross-check exposes the coupling between batch size and learning rate. At lr 2e-3 and 28.9M targets, batches of 8, 4 and 2 give validation BPB of 1.572065, 1.564327 and 1.548231, still monotonically decreasing. In other words the "batch bottoms out at 4" result holds only at lr 1e-3; once the rate is raised, smaller batches keep improving, where batch 2 scored 1.622860 at lr 1e-3.

Under a common calibration regime that advantage almost disappears. The optimal temperature for batch 2 with lr 2e-3 is only 1.025 with a calibration gain of 0.000065, because the model is already well calibrated, giving 1.548165; v3 gains 0.007392 from calibration and reaches 1.549498. The gap shrinks from 0.008659 to 0.001333, about 0.13 standard deviations.

A paired replication of batch 2 with lr 2e-3, using an identical configuration, then returned 1.555356 against the original 1.548231, a spread of 0.007125 or about 0.7 standard deviations, for a two-sample mean of 1.551793. The original 0.008659 advantage was therefore mostly a single point fluctuation, and the two-sample mean is still worse than v3 under a common calibration regime (1.551793 against 1.549498). Pushing on to batch 1 degrades

sharply to 1.660200, and batch 2 with a higher rate of  $2.5\text{e-}3$  is also worse at 1.560446, so the small-batch direction is closed.

This report therefore does not issue a new version, and records two methodological rules instead. Two recipes must be compared after each has been given its own optimal temperature, otherwise changes of the "smaller batch and higher rate" kind are systematically overstated because calibration headroom moves the other way. And any single-point difference near the noise floor must be replicated before it is believed: here a single point appeared to win by 0.0087 and the advantage vanished on replication.

## 5.4 Normalisation and feed-forward factorial

Everything above asks how this block should be trained. Explaining why the block itself is RMSNorm plus SwiGLU needs the two factors separated instead of asserted jointly. The factorial below runs on the baseline scale (width 128, depth 4, about 1.08M parameters). All four cells share 9,830,400 processed targets, batch 32, peak learning rate  $1\text{e-}3$ , three percent warmup followed by cosine decay, and seed 17, so a difference between two cells can only come from the component that was swapped.

**Table 11.** Normalisation by feed-forward factorial: four cells at 9.83M targets, batch 32, peak learning rate  $1\text{e-}3$ , seed 17.

| Normalisation | Feed-forward         | Parameters | Validation BPB |
|---------------|----------------------|------------|----------------|
| LayerNorm     | GELU (4x wide)       | 1,083,648  | 2.101695       |
| LayerNorm     | SwiGLU (2.667x wide) | 1,083,136  | 2.045564       |
| RMSNorm       | GELU (4x wide)       | 1,082,496  | 2.095283       |
| RMSNorm       | SwiGLU (2.667x wide) | 1,081,984  | 2.040526       |

**Table 12.** Main effects and interaction of the same factorial

| Effect                    | Definition                                                            | Value     |
|---------------------------|-----------------------------------------------------------------------|-----------|
| Feed-forward main effect  | SwiGLU 2.667x replacing GELU 4x, averaged over both rows              | -0.055444 |
| Normalisation main effect | RMSNorm replacing LayerNorm, averaged over both rows                  | -0.005725 |
| Interaction               | difference between the feed-forward gain under the two normalisations | -0.001373 |

The reading is direct: almost the whole gain comes from the feed-forward network (-0.055444); RMSNorm adds roughly one tenth more (-0.005725); the interaction (-0.001373) is far smaller than either main effect. The two choices are therefore close to additive and can be explained and adopted separately, instead of being presented as a single indivisible "combination mechanism". The four cells differ in parameter count by less than 0.16 percent, so the feed-forward main effect cannot be dismissed as extra parameters: the 2.667x width exists precisely to match the FLOPs of a 4x-wide GELU layer.

The boundary of this evidence has to be stated: the factorial was run only at the baseline scale and was never repeated cell by cell at the final width 320 / depth 8. It supports the choice of RMSNorm plus SwiGLU, but it does not produce a number that transfers exactly to the final scale, and it is used here as directional evidence only.

## 5.5 Position representation and weight sharing

The second group compares position representation and tied output/input weights close to the final scale (width 320, depth 8). Conditions are again 9,830,400 targets, batch 32, peak learning rate  $1\text{e-}3$ , three percent warmup and cosine decay; each position variant is paired across seed 17 and seed 137, so structural gains can be separated from single-run noise.

**Table 13.** Position representation by weight sharing: 9.83M targets, batch 32, peak learning rate  $1\text{e-}3$ .

| Position           | Weights | Parameters | Seed 17  | Seed 137 |
|--------------------|---------|------------|----------|----------|
| learned embeddings | tied    | 10,570,560 | 1.820432 | -        |

| Position           | Weights | Parameters | Seed 17  | Seed 137 |
|--------------------|---------|------------|----------|----------|
| learned embeddings | untied  | 11,225,920 | 1.786678 | -        |
| RoPE               | tied    | 10,488,640 | 1.754858 | 1.741184 |
| RoPE               | untied  | 11,144,000 | 1.745039 | 1.737502 |

**Table 14.** Separating two apparently independent gains

| Comparison                                                  | Delta     | Reading                                          |
|-------------------------------------------------------------|-----------|--------------------------------------------------|
| RoPE replacing learned positions (tied, seed 17)            | -0.065574 | also removes 81,920 position parameters          |
| untied, under learned positions (seed 17)                   | -0.033754 | the largest apparent gain                        |
| untied, under RoPE (seed 17)                                | -0.009819 | only a third of the previous row                 |
| untied, under RoPE (seed 137)                               | -0.003682 | same direction, smaller size                     |
| untied, under RoPE, mean of two seeds                       | -0.006750 | below the single-run noise sigma of about 0.0102 |
| interaction: the untied gain under the two position choices | -0.027004 | most of the untied gain is confounded with RoPE  |

This is the clearest example in the report of why a combined comparison is not a causal claim. Reading only the best cell (RoPE plus untied, 1.737502) invites attributing -0.033754 to untying; once the two rows are differenced, untying on top of RoPE is worth only -0.006750 on average, below the single-run noise sigma of about 0.0102 measured in section 5.1. Untying also costs 655,360 extra parameters, about 2.6 MiB of inference assets, so the final model keeps tied weights. RoPE goes the other way: it gives -0.065574 on seed 17 and -0.079248 on seed 137, consistent in sign and far larger than the noise, while removing the  $256 \times 320 = 81,920$  learned position parameters, so it is adopted.

## 5.6 Training-only mechanisms and the trainer ladder

This subsection answers two questions that are easily conflated: whether training tricks that cost nothing at inference time help, and whether the improvement could in fact come from the trainer rather than from the model.

**Table 15.** Two training-only mechanisms that add no inference-time state (RoPE plus untied, seed 17, 9.83M targets)

| Configuration                                              | Validation BPB | Delta     | Verdict                                     |
|------------------------------------------------------------|----------------|-----------|---------------------------------------------|
| control, same configuration, nothing added                 | 1.745039       | -         | reference                                   |
| tail weight averaging, EMA 0.99 starting at 80 percent     | 1.734143       | -0.010896 | single run, below the replication threshold |
| multi-token prediction, MTP k=1 with auxiliary weight 0.15 | 1.739964       | -0.005075 | same as above                               |

Neither mechanism adds inference-time state or computation: EMA only averages tail weights from one optimisation trajectory, and MTP only adds an auxiliary target during training. Both could therefore be added freely under the rule that training duration is unrestricted, but their gains (-0.010896 and -0.005075) come from single runs and neither exceeds the single-run noise sigma of about 0.0102 measured in section 5.1, so they did not enter the frozen recipe. One boundary has to be recorded: neither mechanism was ever re-measured under the v3 recipe (batch 4, peak learning rate  $1.5e-3$ , 28.9M targets). This report claims neither that they hold at that setting nor that they fail there.

**Table 16.** Trainer changes applied cumulatively to the supplied baseline model (width 128, depth 4; seed 17; 9.83M targets)

| Cumulative step                                                         | Validation BPB | Delta from previous step |
|-------------------------------------------------------------------------|----------------|--------------------------|
| 1. supplied recipe: with replacement, beta2 0.999, baseline LR schedule | 2.072376       | -                        |

| Cumulative step                               | Validation BPB | Delta from previous step |
|-----------------------------------------------|----------------|--------------------------|
| 2. plus without-replacement window sampling   | 2.071598       | -0.000778                |
| 3. plus AdamW beta2 = 0.95                    | 2.103114       | +0.031516                |
| 4. plus three percent warmup and cosine decay | 2.098476       | -0.004638                |
| net effect of step 1 to step 4                | -              | +0.026100                |

This ladder exists to close the alternative explanation that the improvement comes from the trainer. After every trainer setting used in this project is applied to the same baseline model, validation BPB does not improve: it gets worse by +0.026100. The gains attributed to the model therefore cannot come from sampling, beta2 or the learning-rate schedule; they have to come from the structure, discussed in section 3 and section 5.4. This is also a negative result worth keeping: moving beta2 from 0.999 to 0.95 is harmful on its own at this budget (+0.031516), and it is retained only because combined with warmup and cosine decay on deeper, wider and longer-trained models it no longer shows a penalty. Every row of the ladder is confounded with the rows above it, so no single row is a causal contribution of one mechanism.

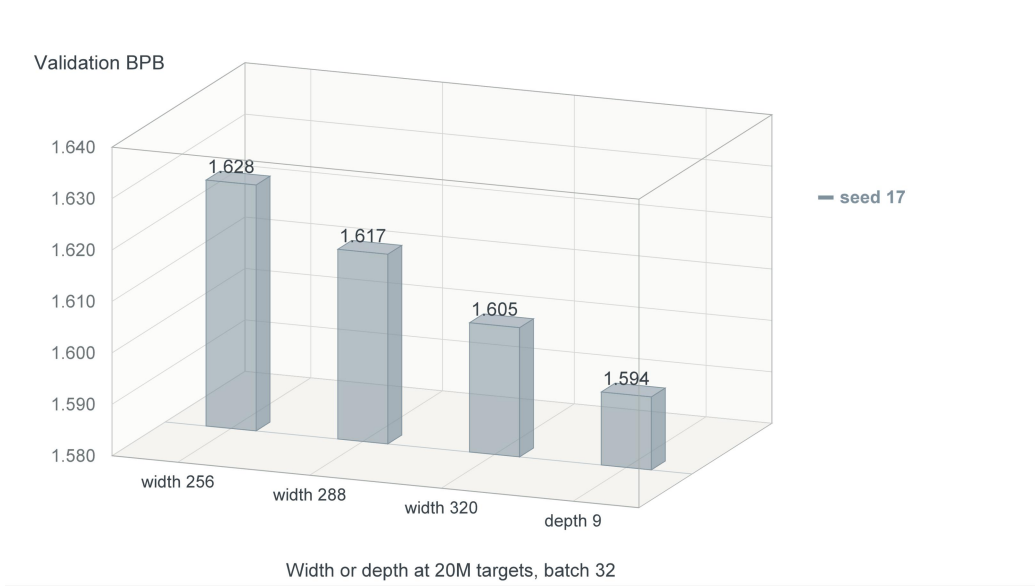
**Table 17.** The earlier learning-rate sweep: batch 32, seed 17, 9.83M targets, learned positions with tied weights

| Peak learning rate | Validation BPB | Relative to 1e-3 |
|--------------------|----------------|------------------|
| 5.0e-4             | 1.887329       | +0.066898        |
| 7.5e-4             | 1.826578       | +0.006146        |
| 1.0e-3             | 1.820432       | +0.000000        |
| 1.25e-3            | 1.850677       | +0.030245        |
| 2.0e-3             | 1.995888       | +0.175457        |
| 4.0e-3             | 1.972859       | +0.152427        |

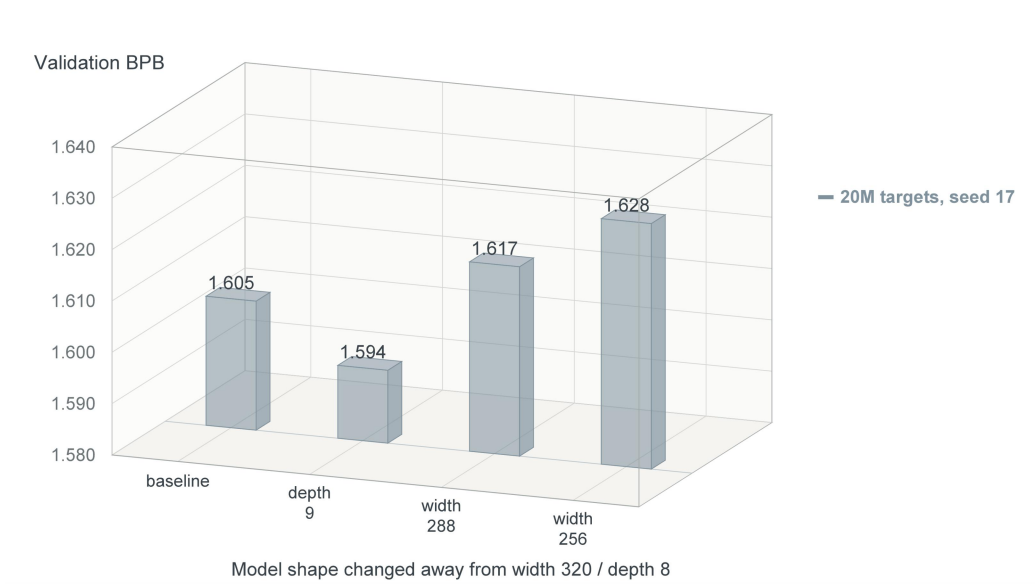
This is the source of the methodological point made in section 5.3: at batch 32 the swept optimum really was 1e-3, and 2e-3 collapsed (+0.175457). Once the batch drops to 4 the same conclusion stops holding and the optimum moves past 1.5e-3. A learning rate has no optimum that is independent of batch size; every change of batch requires a fresh sweep.

## 5.7 Why capacity stops at width 320 / depth 8

Changing width or depth at a fixed token budget shows widths of 256 and 288 to be worse than 320 by 0.022691 and 0.011753, while depth 9 is nominally better by 0.010936 but only half the noise floor and therefore unresolved. The intuition that a smaller model would overfit less does not hold here, and width 320 with depth 8 is the sweet spot for this budget.



**Fig. 10.** Changing width or depth at a fixed token budget. Shrinking the model clearly hurts; deepening it gains nothing outside the noise band.



**Fig. 11.** The paired capacity results: widths 288 and 256 are worse than width 320 by 0.011753 and 0.022691 BPB.

**Table 18.** Capacity, quality and training cost (seed 17, 20M targets, batch 32)

| Configuration                      | Validation BPB | Relative to w320 | Parameters | Verdict    |
|------------------------------------|----------------|------------------|------------|------------|
| width 256 / depth 8                | 1.627825       | +0.022691        | 6,816,000  | harmful    |
| width 288 / depth 8                | 1.616887       | +0.011753        | 8,557,344  | harmful    |
| width 320 / depth 8<br>(reference) | 1.605134       | —                | 10,488,640 | reference  |
| width 320 / depth 9                | 1.594198       | -0.010937        | 11,717,760 | unresolved |

Two further capacity directions were excluded by the resource limits rather than by experiment. Width 384 would push the CPU time ratio to between 4.5 and 5.0x, too close to the hard limit to leave any re-measurement margin, and an earlier short-budget comparison found both width 384 and depth 10 clearly worse. Within this evaluation budget, capacity exploration is effectively closed.

## 6 Final Results and Resources After Freezing

The v3 freeze manifest was written before the test run with its test-result field empty; the same checkpoint was then scored once. Every hash was checked against the manifest.

**Table 19.** Final results on the complete test split

| Version                | Validation BPB | Complete test BPB | Token PPL | Relative to baseline |
|------------------------|----------------|-------------------|-----------|----------------------|
| Course baseline        | 2.071087       | 2.101265          | 80.848    | —                    |
| Frozen v1              | 1.622441       | 1.646152          | 31.83     | -21.66%              |
| Frozen v2              | 1.588095       | 1.611178          | 29.023    | -23.32%              |
| Frozen v3 (this round) | 1.556890       | 1.571349          | 26.704    | -25.22%              |

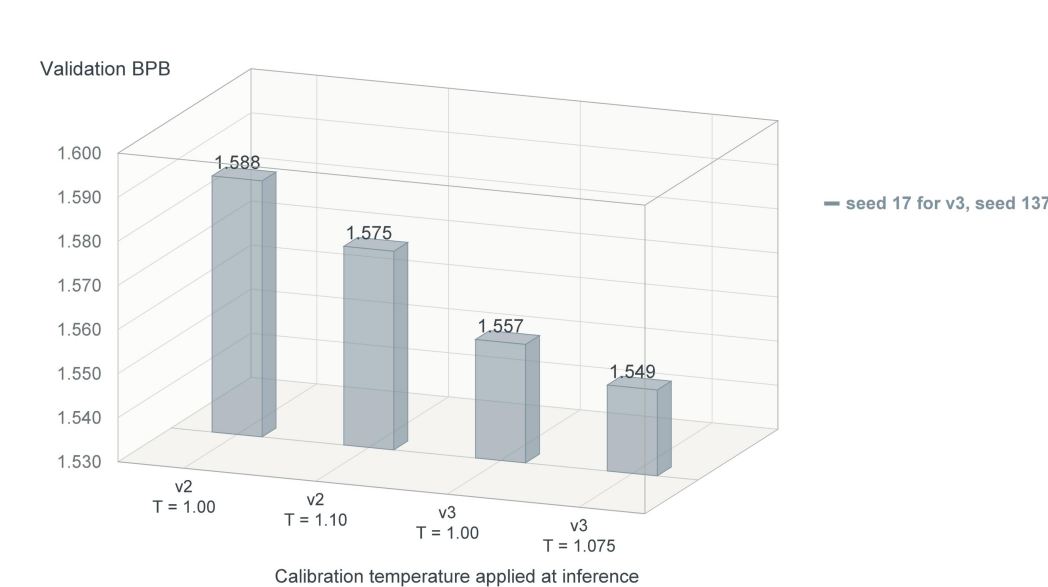
### 6.1 Training and search costs

The disclosed search totals 32 controlled runs and 773,704,448 processed targets this round. The freeze manifest records the snapshot as it stood at freeze time (24 runs and 542,427,904 targets); the later learning-rate runs are counted here as well. On top of that, 33 runs and 429,047,808 targets went into v1 and v2 development. A single 20M-target run takes about 16 to 20 minutes on MPS bf16, and the final recipe (28.9M targets at batch 4) about 27 minutes. Training duration and search size are unlimited under the course rules and only need disclosure.

This round raises the optimisation-step count by about eightfold while leaving the parameter count unchanged; inference adds only one temperature division over the logits, whose compute and storage costs are negligible beside the Transformer forward pass.

### 6.2 What the test improvement corresponds to

v3 test BPB is 0.039830 below v2, but the result is not an additive causal decomposition: the 0.031205 uncalibrated validation gap combines changes in seed, batch, learning rate, decay floor and checkpoint location, whereas only the 0.007392 temperature gain is a paired comparison on the same v3 checkpoint. The architecture and parameter count are unchanged, and the inference graph explicitly adds the temperature division.



**Fig. 12.** Validation BPB before and after temperature scaling. The v2 value  $T = 1.10$  is a post-freeze diagnostic and official v2 remains at  $T = 1.00$ ; v3 selected  $T = 1.075$  before its freeze.

**Table 20.** Calibration and the half-split generalisation check

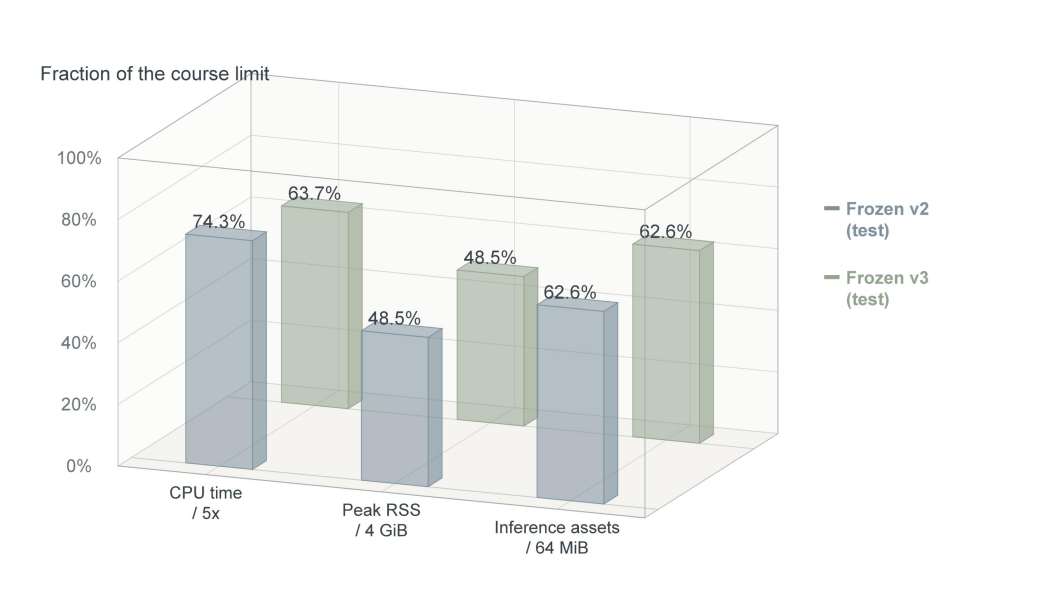
| Item                        | Value | Note                                          |
|-----------------------------|-------|-----------------------------------------------|
| Diagnostic optimum T for v2 | 1.100 | post-freeze only; official v2 uses $T = 1.00$ |
| Optimal T for v3            | 1.075 | validation BPB 1.556890 to 1.549498           |

| Item                      | Value      | Note                                                     |
|---------------------------|------------|----------------------------------------------------------|
| Gain on the fitting half  | 0.012501   | T fitted on the even half only                           |
| Gain on the held-out half | 0.013546   | scored on the odd half; no smaller than the fitting half |
| Inference cost            | negligible | no extra parameters; one scalar division, assets +1 KB   |

One interaction is easy to miss: smaller batches carry more training noise and leave less calibration headroom (0.013023 for v2 against 0.007392 for v3). The batch-4 gain and the calibration gain therefore cannot be added and compared against batch 32; both must be compared under the same calibration regime, which is how this report treats them.

### 6.3 Resource thresholds and measurement protocols

Resource measurement uses a median of five repetitions. Each repetition loads one checkpoint in a fresh process and scores the complete test split in CPU FP32, with the baseline measured in the same invocation so the ratio never depends on a stale hard-coded number.



**Fig. 13.** Resource use of v2 and v3 on the test split, as a fraction of each limit. Both predictors remain below all three bounds, with nearly identical asset use.

**Table 21.** Test resource measurement (median of five, FP32, four threads)

| Metric           | v2 measured | v3 measured | Limit  | Verdict |
|------------------|-------------|-------------|--------|---------|
| CPU time ratio   | 3.713x      | 3.187x      | 5x     | pass    |
| Peak RSS         | 1.940 GiB   | 1.941 GiB   | 4 GiB  | pass    |
| Inference assets | 40.033 MiB  | 40.034 MiB  | 64 MiB | pass    |

The compute limit is not the binding constraint because fixed overhead and the inefficiency of the reference implementation take a large share: the course baseline has only 1.09M parameters and its small matrix multiplications run far below the efficiency of a width-320 model, so scaling parameters by nearly ten raises the time ratio only to about three. This report also records that the ratio itself is unstable across sessions, so freezing always re-measures instead of trusting a single number.

## 7. How the supplementary ablation changes the subsequent direction

This section collects every direction that was rejected or left unresolved. They matter as much as the positive results: these negative results closed the three conventional routes of more regularisation, more capacity and more training, which left batch size and learning rate as the only remaining levers that cost nothing at inference.

**Table 22.** Every ablation of this round and its verdict

| Direction             | Validation BPB | Relative  | Verdict    | Consequence                         |
|-----------------------|----------------|-----------|------------|-------------------------------------|
| droppath 0.1          | 1.650154       | +0.045020 | harmful    | training-time regularisation closed |
| EMA 0.999             | 1.610504       | +0.005370 | harmful    | weight averaging not adopted        |
| width 256             | 1.627825       | +0.022691 | harmful    | shrinking capacity is not viable    |
| width 288             | 1.616887       | +0.011753 | harmful    | shrinking capacity is not viable    |
| depth 9               | 1.594198       | -0.010937 | unresolved | needs paired replication            |
| token dropout 0.05    | 1.602067       | -0.003067 | unresolved | gain below the noise floor          |
| embedding dropout 0.1 | 1.604153       | -0.000981 | unresolved | gain below the noise floor          |

### 7.1 Why regularisation fails here

Regularisation is negative or unresolved across the board, which runs against the common expectation that a small dataset wants more of it. The reason is that the limiting factor here is not how often the model has seen the training set, but that the training set itself holds only 3.6M tokens. In that regime the model has not exhausted its fitting capacity, and any perturbation, whether dropping branches, tokens, dimensions or averaging weights, removes signal it could otherwise have learned. The training-budget sweep in section 5 shows the same limit from another direction: beyond 20M targets no stable gain exceeds run variability.

### 7.2 Directions excluded by the resource limits

Width 384 with depth 8 still fits the 64 MiB asset limit (about 57 MiB) but pushes the CPU time ratio to between 4.5 and 5.0x, too close to the hard limit given the instability this report records. That direction is judged infeasible rather than refuted. Batches of 1 already rebounded in the measurement and need no further work.

## 8 Three Seeds and Best-Checkpoint Selection

v2 was selected from three seeds on validation (seed 137 scored 1.588095, seed 17 scored 1.600916 and seed 233 scored 1.611337, mean 1.600116). After establishing the noise floor this round, that three-seed selection advantage turns out to be only 0.012, the same order as the between-run standard deviation of 0.010, so a substantial part of it was selecting noise.

**Table 23.** Paired replications against the v2 three-seed spread

| Source             | Content                                    | Mean     | Standard deviation |
|--------------------|--------------------------------------------|----------|--------------------|
| This round, paired | three runs of one configuration (batch 32) | 1.613278 | 0.010223           |
| v2 three seeds     | seeds 137 / 17 / 233                       | 1.600116 | 0.011642           |

### 8.1 How non-reproducible training changed the decision rules

Three rules follow and were applied to every conclusion here. A single-run difference below 0.02 is not interpreted. An adopted mechanism must show a monotone trend or a paired replication. And when selecting checkpoints or seeds, the mean and the spread are reported rather than only the best run. These rules are what

forced the downgrade of the two earlier conclusions about the overfitting turning point and the scope of the learning-rate sweep.

## 8.2 Legitimacy boundary of the calibration temperature

Temperature calibration can be misread as tuning for a score, so its boundary is stated explicitly. It does not change the model's discriminative ability, only the calibration of its output distribution. The output stays normalised, strictly causal and stateless, and the scorer's normalization check still passes. T is a hyperparameter selected on validation, of the same nature as a learning rate or weight decay, and no test information was used. It also passed a half-split generalisation check, in which a temperature fitted on one half of validation gains no less on the other half. Both this report and the freeze manifest state plainly that the temperature was fitted on validation.

## 9 Correctness and submission requirements

### 9.1 Reproducing materials

The freeze manifest records the SHA-256 of the checkpoint, student.py, evaluate.py and the tokenizer. Training uses train\_student.py and scoring uses the supplied evaluate.py unmodified. Per-run configuration, seed, training seconds, parameter count and checkpoint hash live in each run directory's metrics.json and are summarised in code/results/grid.tsv.

**Table 24.** Freeze record (v3)

| Item                               | Value                                                            |
|------------------------------------|------------------------------------------------------------------|
| Frozen at                          | 2026-09-26T12:53:30+08:00                                        |
| Checkpoint                         | runs/2026-09-26_phase8/p8_bs4_lr1500/checkpoint_calibrated.pt    |
| Checkpoint SHA-256                 | ba65ec2e216ca5054e9624efa6577e1af9ce14944ced39a4c2f89c4c6acd65d7 |
| student.py SHA-256                 | 6888d7dfc53cd0ae1470d8842591c76025aced8620367b86c43b772334cbfcd6 |
| evaluate.py SHA-256                | 128bcb2dab0be0d427505bddb4671e3ab3a8f78e114be79a689c0f9029af133d |
| Tokenizer SHA-256                  | 020d1bc6aa4449c4f352b2e03d0e0fb4f39287f15297705e421b1fa7d817262e |
| Parameters                         | 10,488,640                                                       |
| Test BPB (single run after freeze) | 1.571349                                                         |

### 9.2 Assignment requirement coverage

**Table 25.** Course requirements and where the evidence lives

| Course requirement                            | Where it is covered                                |
|-----------------------------------------------|----------------------------------------------------|
| Initial supplied baseline                     | section 4, Table 5 and Fig. 2                      |
| Comparison at equal processed targets         | section 5.3, Table 9 (all at 20M targets)          |
| Ablation of the key mechanism                 | sections 5.2, 5.3, 5.4 and section 7               |
| Validation-based development and selection    | sections 2.1 and 8                                 |
| Prediction quality against computational cost | sections 5.3 and 6.3, including the resource table |
| Training costs, seeds and checkpoint ancestry | section 6.1, section 8 and Appendix A              |
| The 5x CPU / 4 GiB / 64 MiB limits            | section 6.3 resource table and figure              |
| One test evaluation after freezing            | section 6 Table 12 and section 9.1 Table 17        |

## 10 Conclusions, Limitations, and AI Disclosure

### 10.1 Conclusion

Under the fixed WikiText-2 text, BPE-2048 tokenizer, 256-token context and CPU FP32 scorer, frozen v3 reaches a complete test BPB of 1.571349, 25.22% below the course baseline and 0.039830 below the previously frozen v2. All three resource limits pass, the three resource limits pass; the parameter count matches v2, while temperature scaling adds one scalar division and about 1 KB to the inference assets.

The strongest methodological conclusion is that a batch change requires a fresh learning-rate sweep and that recipes must be compared under a common calibration protocol. The uncalibrated v3 validation score is 0.031 below frozen v2, but this joint gap also includes the seed, decay floor and checkpoint change and cannot be assigned to batch size and learning rate alone; temperature scaling supplies a separate paired gain of 0.007392 at the cost of one scalar division.

### 10.2 Limitations

First, non-reproducible training is the dominant methodological constraint. Three runs of one configuration have a standard deviation of 0.010, so no single comparison counts unless it is backed by a paired replication or a monotone trend, and two earlier conclusions were downgraded on that basis.

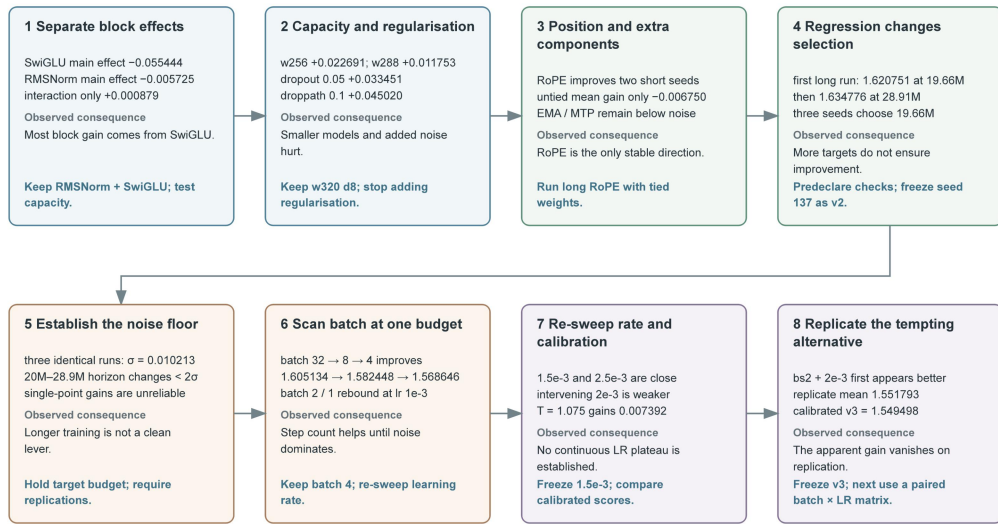
Second, batch 4 has one observation under the common 20M-target comparison, while the three batch-32 replications use 28.91M targets and estimate variability rather than a matched effect. A follow-up should repeat batches 32, 8 and 4 at least twice under the same budget and re-sweep the learning rate within each batch. The isolated 1.5e-3 and 2.5e-3 runs are close, but the weaker 2e-3 run between them prevents a continuous-plateau claim.

Third, the architecture and parameter count remain fixed, while the result combines a training and checkpoint recipe change with inference-time temperature scaling. A stronger experiment would preregister a paired 28.91M-target matrix, report both uncalibrated and independently calibrated validation BPB, and use means with confidence intervals to resolve the interactions among batch size, learning rate and decay floor before freezing another version.

Fourth, the calibration temperature improves calibration rather than discriminative ability. Although it passed the half-split check, its origin must be stated plainly and it must not be presented as an architectural improvement.

### 10.3 AI assistance disclosure

The experiment scripts, figure generation and report typesetting were produced with AI assistance, including the implementation of three training-only regularisation knobs (droppath, token dropout and embedding dropout), the calibration and freeze-manifest tooling, and the figure and Word generation pipeline. All experimental design, metric decisions and conclusions were confirmed by the author. No AI component took part in any selection on the test split, and no training data beyond the supplied text was used.



**Fig. 14.** The debugging path from v1/v2 structure, capacity, position and checkpoint selection to the v3 noise, batch, learning-rate and calibration checks; each observation directly determines the next experiment.

## References

1. DASE7506 Project 1 course guide (GUIDE.md). 2. Supplied baseline implementation and scorer (model.py / evaluate.py / common.py, unmodified). 3. Vaswani et al., “Attention Is All You Need,” NeurIPS, 2017. 4. Zhang and Sennrich, “Root Mean Square Layer Normalization,” NeurIPS, 2019. 5. Shazeer, “GLU Variants Improve Transformer,” arXiv:2002.05202, 2020. 6. Su et al., “RoFormer: Enhanced Transformer with Rotary Position Embedding,” Neurocomputing, 2024. 7. Loshchilov and Hutter, “Decoupled Weight Decay Regularization,” ICLR, 2019. 8. Merity et al., “Pointer Sentinel Mixture Models,” ICLR, 2017. 9. This project's v1 and v2 technical reports and experiment index (code/results/ and run directories).

## Appendix A Full index of saved experiments

Each row comes from the metrics.json of that run directory.

### A.0 Earlier structural and representation controls (v1 / v2 stage)

#### A.0. v1 / v2 阶段的历史运行

| Run                       | Seed | Batch | Peak LR | Targets | Validation BPB |
|---------------------------|------|-------|---------|---------|----------------|
| recipe_exact_baseline_s17 | 17   | 32    | 0.001   | 9.83M   | 2.072376       |
| recipe_norepl_s17         | 17   | 32    | 0.001   | 9.83M   | 2.071598       |
| recipe_norepl_beta095_s17 | 17   | 32    | 0.001   | 9.83M   | 2.103114       |
| recipe_full_student_s17   | 17   | 32    | 0.001   | 9.83M   | 2.098476       |
| s0_lr1e3                  | 17   | 32    | 0.001   | 9.83M   | 2.101695       |
| ablation_ln_swiglu_s17    | 17   | 32    | 0.001   | 9.83M   | 2.045564       |
| ablation_rms_gelu_s17     | 17   | 32    | 0.001   | 9.83M   | 2.095283       |
| s_lr1e3                   | 17   | 32    | 0.001   | 9.83M   | 2.040526       |
| lr1e3_a                   | 17   | 32    | 0.001   | 9.83M   | 1.820432       |
| ablation_untied_s17       | 17   | 32    | 0.001   | 9.83M   | 1.786678       |
| ablation_rope_s17         | 17   | 32    | 0.001   | 9.83M   | 1.754858       |
| ablation_rope_s137        | 137  | 32    | 0.001   | 9.83M   | 1.741184       |

| Run                             | Seed | Batch | Peak LR | Targets | Validation BPB |
|---------------------------------|------|-------|---------|---------|----------------|
| ablation_rope_untied_s17        | 17   | 32    | 0.001   | 9.83M   | 1.745039       |
| ablation_rope_untied_s137       | 137  | 32    | 0.001   | 9.83M   | 1.737502       |
| ablation_rope_untied_ema099_s17 | 17   | 32    | 0.001   | 9.83M   | 1.734143       |
| ablation_rope_untied_mtp1_s17   | 17   | 32    | 0.001   | 9.83M   | 1.739964       |
| drop005_lr1e3_a                 | 17   | 32    | 0.001   | 9.83M   | 1.853883       |
| wd2e1_lr1e3_a                   | 17   | 32    | 0.001   | 9.83M   | 1.820517       |
| lr5e4_a                         | 17   | 32    | 0.0005  | 9.83M   | 1.887329       |
| lr7p5e4_a                       | 17   | 32    | 0.00075 | 9.83M   | 1.826578       |
| lr1p25e3_a                      | 17   | 32    | 0.00125 | 9.83M   | 1.850677       |
| lr2e3_a                         | 17   | 32    | 0.002   | 9.83M   | 1.995888       |
| lr4e3_a                         | 17   | 32    | 0.004   | 9.83M   | 1.972859       |
| capacity_w320d10_s17            | 17   | 32    | 0.001   | 9.83M   | 1.872762       |
| capacity_w384d8_s17             | 17   | 32    | 0.001   | 9.83M   | 1.829156       |

## A.1 Training budget and full anneal

A.1. Training budget and full anneal (2026-09-26\_phase1)

| Run     | Seed | Batch | Peak LR | Targets | Validation BPB |
|---------|------|-------|---------|---------|----------------|
| h16     | 17   | 32    | 0.001   | 16.01M  | 1.640587       |
| h20     | 17   | 32    | 0.001   | 20.00M  | 1.605134       |
| h20_ema | 17   | 32    | 0.001   | 20.00M  | 1.610504       |
| h24     | 17   | 32    | 0.001   | 24.00M  | 1.605983       |
| h28p9   | 17   | 32    | 0.001   | 28.91M  | 1.603377       |

## A.2 Regularisation ablation

A.2. Regularisation ablation (2026-09-26\_phase2)

| Run        | Seed | Batch | Peak LR | Targets | Validation BPB |
|------------|------|-------|---------|---------|----------------|
| p2_all     | 17   | 32    | 0.001   | 20.00M  | 1.661331       |
| p2_all_ema | 17   | 32    | 0.001   | 20.00M  | 1.656461       |
| p2_dp10    | 17   | 32    | 0.001   | 20.00M  | 1.650154       |
| p2_emb10   | 17   | 32    | 0.001   | 20.00M  | 1.604153       |
| p2_td05    | 17   | 32    | 0.001   | 20.00M  | 1.602067       |

## A.3 Capacity and batch size, first batch

A.3. Capacity and batch size, first batch (2026-09-26\_phase3)

| Run       | Seed | Batch | Peak LR | Targets | Validation BPB |
|-----------|------|-------|---------|---------|----------------|
| p3_bs16   | 17   | 16    | 0.001   | 20.00M  | 1.594155       |
| p3_bs8    | 17   | 8     | 0.001   | 20.00M  | 1.578192       |
| p3_w256d8 | 17   | 32    | 0.001   | 20.00M  | 1.627825       |

| Run       | Seed | Batch | Peak LR | Targets | Validation BPB |
|-----------|------|-------|---------|---------|----------------|
| p3_w288d8 | 17   | 32    | 0.001   | 20.00M  | 1.616887       |
| p3_w320d9 | 17   | 32    | 0.001   | 20.00M  | 1.594198       |

#### A.4 Batch size, second batch, and paired replications

A.4. Batch size, second batch, and paired replications (2026-09-26\_phase7)

| Run        | Seed | Batch | Peak LR | Targets | Validation BPB |
|------------|------|-------|---------|---------|----------------|
| p7_bs4     | 17   | 4     | 0.001   | 20.00M  | 1.568646       |
| p7_bs4_h28 | 17   | 4     | 0.001   | 28.91M  | 1.568068       |
| p7_bs8_h28 | 17   | 8     | 0.001   | 28.91M  | 1.578710       |
| p7_rep1    | 17   | 32    | 0.001   | 28.91M  | 1.612661       |
| p7_rep2    | 17   | 32    | 0.001   | 28.91M  | 1.623795       |

#### A.5 Learning-rate re-sweep and batch bounds

A.5. Learning-rate re-sweep and batch bounds (2026-09-26\_phase8)

| Run           | Seed | Batch | Peak LR | Targets | Validation BPB |
|---------------|------|-------|---------|---------|----------------|
| p8_bs1        | 17   | 1     | 0.001   | 20.00M  | 1.660514       |
| p8_bs2        | 17   | 2     | 0.001   | 20.00M  | 1.622860       |
| p8_bs4_lr1500 | 17   | 4     | 0.0015  | 28.91M  | 1.556890       |
| p8_bs4_lr750  | 17   | 4     | 0.00075 | 28.91M  | 1.596950       |

#### A.6 Learning-rate upper sweep and batch check

A.6. Learning-rate upper sweep and batch check (2026-09-26\_phase10)

| Run           | Seed | Batch | Peak LR | Targets | Validation BPB |
|---------------|------|-------|---------|---------|----------------|
| p10_bs2_lr2e3 | 17   | 2     | 0.002   | 28.91M  | 1.548231       |
| p10_bs8_lr2e3 | 17   | 8     | 0.002   | 28.91M  | 1.572065       |
| p10_lr25e3    | 17   | 4     | 0.0025  | 28.91M  | 1.557149       |
| p10_lr2e3     | 17   | 4     | 0.002   | 28.91M  | 1.564327       |
| p10_lr3e3     | 17   | 4     | 0.003   | 28.91M  | 1.564697       |

Each run directory also keeps the config, seed, training seconds, parameter count and checkpoint SHA-256. Historical v1 and v2 runs are indexed in code/results/grid.tsv.